

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics
Year 8

WORKBOOK

Probability

Year 8 Mathematics • Workbook

Name _____ Class _____ Teacher _____

How this workbook works

Each lesson is **three pages**. Copy your teacher's **notes**, then work through **We do** (dark boxes: we solve these together; copy every step) and **You do** (your turn, one round at a time). Rate each success criterion at the end.

Page 1 of each lesson is **Easy** ●, page 2 is **Medium** ●● and page 3 is **Hard** ●●●: a harder example, mixed questions (a tag shows an earlier lesson or year) and worded problems. A question tagged **Extension** uses Stage 5 wording.

NSW Mathematics K–10 Syllabus (2022), Stage 4: MA4-PRO-C-01. We acknowledge the Bundjalung people, the Traditional Custodians of the land on which we learn.

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Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

1 Chance and the sample space

Probability • MA4-PRO-C-01 **EASY •**

LEARNING INTENTION

We are learning to describe chance with words and with numbers from 0 to 1, list a sample space, and decide whether outcomes are equally likely.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* place an event on the probability scale from 0 to 1. ○○○
- *I can* list the sample space of a chance experiment. ○○○
- *I can* decide, with a reason, whether outcomes are equally likely. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Answer each part. Give a reason where you are asked to.

a The scale

Mark: A the sun rises tomorrow; B rolling a 7 on a die; C a coin lands heads.



b Sample space

List the sample space. How many outcomes are there?



c Equally likely?

Is each colour equally likely? Explain.



d Sample space

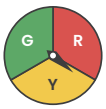
List the sample space for the day of the week a person was born on. How many outcomes?

YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER impossible = 0 · unlikely: between 0 and $\frac{1}{2}$ · even chance = $\frac{1}{2}$ · likely: between $\frac{1}{2}$ and 1 · certain = 1

ROUND 1 Questions 1–4: count the blue sectors and the total sectors, then choose a chance word for landing on blue.

1



Blue _____
Total _____
Word _____

2



Blue _____
Total _____
Word _____

3



Blue _____
Total _____
Word _____

4



Blue _____
Total _____
Word _____

ROUND 2 Questions 5–8: list the sample space in braces { }, then count the outcomes.

5 Tossing a coin



Sample space { _____ }
Outcomes _____

6 Rolling a die



Sample space { _____ }
Outcomes _____

7 Spinning this spinner



Sample space { _____ }
Outcomes _____

8 Choosing one tile



Sample space { _____ }
Outcomes _____

1 Chance and the sample space

Probability • MA4-PRO-C-01

MEDIUM ●●

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER sample space: every possible outcome, each written once, in braces { } · event: the outcomes we want · equally likely: the same chance (equal sectors, a fair die, chosen at random)

ROUND 3 Questions 9–11: list the sample space. Write each outcome only once, even if it repeats.

9 This spinner is spun once.



Sample space { }

Outcomes

10 The first letter of the month a person was born in.

Sample space { }

Outcomes

11 The result of a netball game for Kingscliff.

Sample space { }

Outcomes

ROUND 4 Questions 12–14: list the outcomes in each event, in braces { }.

12 Rolling an odd number on a die.



Event { }

13 Choosing a vowel from these tiles.

C U D G E N

Event { }

14 A house number greater than 4 but less than 10.

Event { }

ROUND 5 Questions 15–17: are the outcomes equally likely? Write *yes* or *no*, then give a reason.

15 Spinning this spinner.



Yes / no

Reason

16 Rolling a fair die.



Yes / no

Reason

17 Passing or failing a driving test.

Yes / no

Reason

ROUND 6 Questions 18–20: write the chance word for each probability: *impossible, unlikely, even chance, likely* or *certain*.

18 A probability of **0.9**

Word

19 A probability of $\frac{1}{2}$

Word

20 A **2%** chance

Word

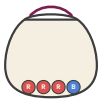
1 Chance and the sample space

Probability • MA4-PRO-C-01 HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Answer in a full sentence, and give the reason.

a Equally likely?
One counter is taken. List the sample space. Are the outcomes equally likely?



b A match
Kingscliff plays Cabarita in a final. List the sample space. Are the outcomes equally likely?

c Order
Least to most likely: A an even number, B a 6, C less than 6, D a 9.



d Forecast
A 30% chance of rain: is rain likely or unlikely? Could it still rain?



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER 0 = impossible · $\frac{1}{2}$ = even chance · 1 = certain · list each outcome once · equally likely means the same chance

1 Mark on the scale: A rolling a 6 on a die, B rolling a number less than 7, C a coin landing heads.



2 Write $\frac{3}{4}$ as a decimal and as a percentage. Year 7

Answer

3 One tile is chosen at random. List the sample space.



Answer

4 This spinner is spun. Which is more likely: red or blue? Explain.



Answer

5 Sam says, 'On a die, a 6 is less likely than a 1 because 6 is a bigger number.' Is Sam right? Explain.

Answer

6 A coin lands tails 7 times in a row. Is heads more likely on the next toss? Explain.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1 The forecast for the Tweed is 'a 70% chance of rain'. Describe the chance in words. Is rain certain? Explain.

Answer

2 Colour this spinner's 4 equal sectors so that landing on blue is *likely* but not *certain*. Explain why your spinner works.



Answer

3 Amy and Cara are in 8K; Ben and Dev are in 8R. One of the four is chosen at random as captain. List the sample space. Is the captain more likely to come from 8K?

Answer

2 Probability of an event

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to find the probability of an event when the outcomes are equally likely, and write it as a fraction, decimal and percentage.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* count the favourable outcomes and the total outcomes. ○○○
- ▸ *I can* use P(E) = favourable ÷ total, and simplify. ○○○
- ▸ *I can* write a probability as a fraction, a decimal and a percentage. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Write the fraction in simplest form.

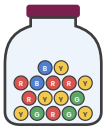
a Spinner: P(red)



b Die: P(an odd number)



c Lollies: P(yellow or blue)



d Tiles: P(a vowel)



YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER $P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ · simplify: divide the top and bottom by the same number · a die: {1, 2, 3, 4, 5, 6}

ROUND 1 Questions 1–4: count the blue sectors and the total sectors, then write P(blue) in simplest form.

1



Blue _____
Total _____
P(blue) _____

2



Blue _____
Total _____
P(blue) _____

3



Blue _____
Total _____
P(blue) _____

4



Blue _____
Total _____
P(blue) _____

ROUND 2 Questions 5–8: a fair die is rolled once. List the favourable outcomes, then find the probability.

5 Find P(rolling a 5).



Favourable { } _____
P = _____

6 Find P(an even number).



Favourable { } _____
P = _____

7 Find P(a number less than 3).



Favourable { } _____
P = _____

8 Find P(a factor of 6).



Favourable { } _____
P = _____

2 Probability of an event

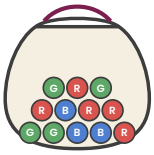
Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER count every counter, lolly or card · 'or' means add the counts · decimal: top ÷ bottom · percentage: × 100%

ROUND 3 — Questions 9–11: one counter is taken at random from this bag. Find each probability.



5 red, 3 blue and 4 green counters.
Count them!

9 Find P(red).

Favourable _____

Total _____

P = _____

10 Find P(blue).

Favourable _____

Total _____

P = _____

11 Find P(red or green).

Favourable _____

Total _____

P = _____

ROUND 4 — Questions 12–14: write each probability as a fraction, a decimal and a percentage.

12 A coin is tossed. P(tails)



Fraction _____

Decimal _____

Percentage _____

13 This spinner is spun. P(red)



Fraction _____

Decimal _____

Percentage _____

14 One tile is chosen. P(an F)

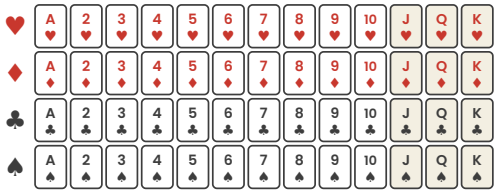


Fraction _____

Decimal _____

Percentage _____

ROUND 5 — Questions 15–17: one card is taken at random from a standard deck. Find each probability.



52 cards: 4 suits of 13. Hearts and diamonds are red. J, Q, K are picture cards (shaded).

15 Find P(an ace).

Favourable _____

P = _____

16 Find P(a heart).

Favourable _____

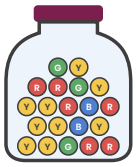
P = _____

17 Find P(a red card).

Favourable _____

P = _____

ROUND 6 — Questions 18–20: one lolly is taken at random from this jar. Give each answer in the form asked.



6 red, 3 green, 9 yellow and 2 blue lollies:
20 in total.

18 P(green), as a decimal

Working _____

Answer _____

19 P(yellow), as a percentage

Working _____

Answer _____

20 P(red or blue), as a fraction

Working _____

Answer _____

2 Probability of an event

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Give each answer in the form asked. Read each question twice.

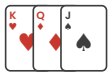
a As a decimal

One coin falls out of the money box. P(gold coin)



b Cards

One card from a 52-card deck. P(a red picture card)



c As a percentage

A die is rolled. P(a number greater than 4)



d Notes

Elena has two $5, four $10 and three $20 notes. She takes one at random. P($20 note)

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER P(E) = favourable ÷ total · picture cards: J, Q, K · answer in the form the question asks for

1 One jelly bean is taken at random. Find P(green or black).



Answer

2 Write $\frac{3}{8}$ as a decimal and as a percentage. Year 7

Answer

3 A whole number from 1 to 15 is chosen at random. Find P(a prime number).



Answer

4 One tile is chosen at random. Find P(an S).



Answer

5 A die is rolled. Find P(rolling at least 3). Extension



Answer

6 500 raffle tickets are sold at the school fete. Mia buys 8. Find P(Mia wins first prize), as a percentage.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 An Esky has 8 lemonade, 5 orange and 2 lime cans. Find P(lemonade). How many cola cans must be added so that P(lemonade) = $\frac{1}{2}$?

Answer

2 Malik has 20 cards: five 5s, four 6s, eight 7s and three 8s. Lara takes one at random. Write P(a 7) as a fraction, decimal and percentage. Then find P(an even number).

Answer

3 The probability that a day in April is rainy is 30%. About how many rainy days are expected in April (30 days)?



Answer

3 Complementary events

Probability • MA4-PRO-C-01 EASY •

LEARNING INTENTION

We are learning to identify the complement of an event, and use $P(\text{not E}) = 1 - P(E)$ to solve problems.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* write the complement of an event. ○○○
- ▸ *I can* show that $P(E) + P(\text{not E}) = 1$. ○○○
- ▸ *I can* use $P(\text{not E}) = 1 - P(E)$ with fractions, decimals and percentages. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Write each complement, then find the probabilities.

a Complement

Write the complement of rolling a 6 on a die. List its outcomes.



b Cards

A pack has 10 red, 6 yellow and 4 green cards. One is chosen. Find P(yellow) and P(not yellow).

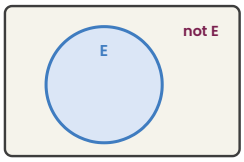
c Decimal

P(rain) = 0.15. Find P(no rain).



d Diagram

Die: E = an even number. Fill in the diagram, then find P(not E).



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER the complement of E is 'not E': every other outcome · $P(E) + P(\text{not E}) = 1$

ROUND 1 Questions 1–4: write the complement of each event.

1 Tossing tails on a coin



Complement _____

2 Rolling a 6 on a die



Complement _____

3 A cloudy day tomorrow

Complement _____

4 Choosing a white chocolate from a box of white and dark chocolates

Complement _____

ROUND 2 Questions 5–8: find P(blue) and P(not blue). Check that they add to 1.

5



P(blue) _____
P(not blue) _____
Check ___ + ___ = 1

6



P(blue) _____
P(not blue) _____
Check ___ + ___ = 1

7



P(blue) _____
P(not blue) _____
Check ___ + ___ = 1

8



P(blue) _____
P(not blue) _____
Check ___ + ___ = 1

3 Complementary events

Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER $P(\text{not E}) = 1 - P(E)$ · percentages: $P(\text{not E}) = 100\% - P(E)$ · decimals: $1 - 0.15 = 0.85$ · fractions: $1 - \frac{1}{4} = \frac{3}{4}$

ROUND 3 — Questions 9–11: a fair die is rolled. Find each probability.

9 Find P(not a 3).



Outcomes { }

P =

10 Find P(not an even number).



Outcomes { }

P =

11 Find P(not a prime number).



Outcomes { }

P =

ROUND 4 — Questions 12–14: use P(not E) = 1 - P(E). Give the answer in the same form as the question.

12 P(rain this weekend) = 85%. Find P(no rain).

Working

Answer

13 P(a carton has a broken egg) = 0.1. Find P(no broken eggs).

Working

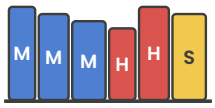
Answer

14 P(choosing a jack from a deck) = $\frac{1}{13}$. Find P(not a jack).

Working

Answer

ROUND 5 — Questions 15–17: one book is chosen at random from this shelf. Find each probability.



3 maths (M), 2 history (H) and 1 sport (S) book.

15 Find P(a history book).

Working

Answer

16 Find P(not a history book).

Working

Answer

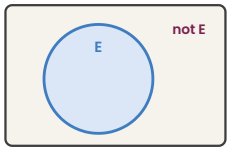
17 Find P(not a science book).

Working

Answer

ROUND 6 — Questions 18–20: show an event and its complement in a diagram, and use the total of 1.

18 A number from 1 to 10 is chosen. E = a multiple of 3. Write the outcomes in E and in not E.



19 For Question 18, find P(E) and P(not E).

Working

Answer

20 Kingscliff's netball team: P(win) = 0.6 and P(draw) = 0.1. Find P(lose).

Working

Answer

3 Complementary events

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Use P(not E) = 1 - P(E). Give each answer in the form asked.

a Percentage
A golfer sinks 74% of putts. Find P(missing a putt).

b Missing value
A bag of toy cars holds only red, blue and white cars. P(red) = 0.4 and P(white) = 0.25. Find P(blue).

c Choose
The complement of 'winning a race' is: A coming last B coming second C not winning D coming 2nd or 3rd

d Raffle
Tahnee buys 5 of the 1000 raffle tickets. Find P(Tahnee does not win).



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER P(not E) = 1 − P(E) · 'not blue or black' means neither blue nor black · all probabilities add to 1

1 A car park has 450 cars, 100 motorbikes and 10 buses. One vehicle is chosen at random. Find P(not a car).

Answer

2 Year 7

Calculate $1 - \frac{3}{8}$, $1 - 0.37$ and $100\% - 64\%$.

Answer

3 A jar has 40 red, 25 blue, 50 black and 15 white jelly beans. One is chosen at random. Find P(not white).

Answer

4 Use the jar in Question 3. Find P(the jelly bean is not blue or black).

Answer

5 A mobile number is chosen at random. Find the decimal probability that its last digit is not 0 or 1.

Answer

6 One tile is chosen at random. Find P(not a vowel).

P R O B A B I L I T Y

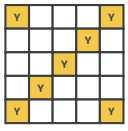
Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 A bag of toy cars holds only red, blue and white cars. P(red) = 0.4 and P(white) = 0.25. Find P(blue). The bag holds 40 cars: how many of each colour are there?

Answer

2 In this game a coin must land in a yellow square of the 5 × 5 grid. Throws that miss the grid are taken again. Find P(winning) and P(not winning).



Answer

3 Sue and Liam are in Year 8; Emily and Matt are in Year 9. One of them is chosen at random as captain. Find P(not Emily), P(the captain is in Year 8) and P(the captain is the teacher).

Answer

4 Probability problems

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to choose and use the right probability skill to solve problems from lists, tables and words.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* decide which skill a problem needs. ○○○
- *I can* set out working with P(E) = favourable ÷ total. ○○○
- *I can* explain my answer in a sentence. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Use the checklist. Write the fraction in simplest form unless asked otherwise.

a A table

28 students on a trip to Japan were born in these countries. One is chosen at random. Find P(born in Asia).

Australia	17
Italy	3
Vietnam	4
Japan	1
China	3

b Letters

One tile is chosen. List the sample space. Is each letter equally likely?

S U C C E S S

c Percentage

Find P(an even number), as a percentage.



d Not

Alex takes one sock from 2 black, 2 blue and 2 red socks. Find P(not red).



YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER P(E) = favourable ÷ total · percentage: × 100% · simplify fractions

ROUND 1 Questions 1–4: a ball is chosen at random from a barrel of balls numbered 1 to 100. Find each probability.

1. Find P(12).



Favourable _____

P = _____

2. Find P(a number greater than 40).



Favourable _____

P = _____

3. Find P(an even number).



Favourable _____

P = _____

4. Find P(a number from 1 to 25).



Favourable _____

P = _____

ROUND 2 Questions 5–8: the spinner has 5 equal sectors numbered 1 to 5. Find each probability as a percentage.

5. Find P(2).



Fraction _____

Percentage _____

6. Find P(an even number).



Fraction _____

Percentage _____

7. Find P(a number less than 5).



Fraction _____

Percentage _____

8. Find P(a number from 1 to 5).



Fraction _____

Percentage _____

4 Probability problems

Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER use the checklist: total → favourable → 'not'? → form asked → check · 'or': add the counts

ROUND 3 — Questions 9–11: one student from the trip is chosen at random. Find each probability.

Born in	Australia	Italy	Vietnam	Japan	China
Students	17	3	4	1	3

28 students went on the school trip to Japan.

9 Find P(born in Australia).

Working _____

Answer _____

10 Find P(not born in Vietnam).

Working _____

Answer _____

11 Find P(born in Italy or China).

Working _____

Answer _____

ROUND 4 — Questions 12–14: Alex takes one sock at random from 2 black, 2 blue and 2 red socks.

12 List the sample space.

Answer _____

13 Find P(a blue sock).

Answer _____

14 Find P(a pink sock).

Answer _____

ROUND 5 — Questions 15–17: a whole number from 1 to 15 is chosen at random. List the favourable outcomes, then find the probability.

15 Find P(an odd number).

Favourable { } _____

P = _____

16 Find P(not a prime number).

Favourable { } _____

P = _____

17 Find P(a factor of 12).

Favourable { } _____

P = _____

ROUND 6 — Questions 18–20: the 7 letters of SUCCESS are written on cards. One card is chosen at random.

S	U	C	C	E	S	S

18 Find P(a vowel).

Working _____

Answer _____

19 Find P(not a C).

Working _____

Answer _____

20 Find P(a letter that is also in the word FAIL).

Working _____

Answer _____

4 Probability problems

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Harder problems: read twice, plan, then solve.

a Small chances

A charity raffle sells 100 000 tickets. Sophia buys 4. Find P(Sophia wins), as a decimal.



b Or

A whole number from 1 to 15 is chosen at random. Find P(a multiple of 3 or 5).



c How many?

Lollies: P(red) = 0.5, P(green) = 0.2 and the rest are purple. Find P(purple). How many of each colour in a bag of 10?

d No memory

A coin shows tails 7 times in a row. Find P(tails on the next toss).



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER $P(\text{not E}) = 1 - P(\text{E})$ · each trial is a fresh start · answer in the form asked

1 A tennis tournament has 34 players from NSW, 20 from Victoria and 6 from Queensland. One is chosen at random. Find P(from NSW).

Answer

2 Use the tournament in Question 1. Find P(not from Victoria).

Answer

3 A raffle sells 5000 tickets and Ben buys 2. Find P(Ben wins), as a decimal.

Answer

4 Write $\frac{7}{15}$ as a decimal, correct to 2 decimal places. Year 7

Answer

5 Find P(not yellow), as a percentage.



Answer

6 A fair die shows a 6 five times in a row. Mai says a 6 is now more likely; Rani says less likely. Who is right?



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 8K has 28 students: 12 catch the bus, 9 walk and the rest are driven to school. One student is chosen at random. Find P(driven) and P(does not walk).

Answer

2 The letters of TWEED HEADS are written on 10 cards. One card is chosen at random. Find P(an E) and P(a vowel).

Answer

3 A charity raffle sells 2500 tickets at $2 each, with one prize. Lucy spends $20 on tickets. Find P(Lucy wins), as a percentage.

Answer

5 Observed probability

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to find observed probabilities from experiments and data, and compare them with theoretical probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* find an observed probability: frequency ÷ total trials. ○○○
- ▸ *I can* tell theoretical and observed probability apart. ○○○
- ▸ *I can* run a simulation with a random number generator. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Observed P(E) = number of times E happened ÷ total trials.

a Observed

A die is rolled 60 times and 2 comes up 15 times. Find the observed P(2).



b Compare

Find the theoretical P(2) for a fair die. Is the observed P(2) higher or lower?

c Spinner

Vinh spins 80 times: red 44, green 25, yellow 11. Find the observed P(red).



d Percentage

Sefina tosses a coin: 135 heads and 115 tails. Find the observed P(tails), as a %.



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER observed $P(E) = \frac{\text{number of times E happened}}{\text{total number of trials}}$ · theoretical: from equally likely outcomes · observed: from results

ROUND 1 Questions 1–4: find each observed probability. Write it in simplest form.

1 A coin is tossed 20 times and lands heads 12 times. Find the observed P(heads).

Working _____

P = _____

2 A die is rolled 30 times and shows 6 four times. Find the observed P(6).

Working _____

P = _____

3 A basketballer scores 35 of 50 shots. Find the observed P(scores).

Working _____

P = _____

4 A spinner is spun 25 times and lands on blue 10 times. Find the observed P(blue).

Working _____

P = _____

ROUND 2 Questions 5–8: is each probability theoretical (T) or observed (O)? Write T or O.

5 $P(\text{heads}) = \frac{1}{2}$, because a coin has two equal sides.



T or O _____

6 Kingscliff won 14 of its 20 games, so P(win) = 0.7.

T or O _____

7 $P(\text{rolling a 3}) = \frac{1}{6}$ for a fair die.



T or O _____

8 120 of last year's 365 days were rainy, so $P(\text{rain}) \approx 0.33$.

T or O _____

5 Observed probability

Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 11 QUESTIONS

REMEMBER the total number of trials goes on the bottom · add the frequencies to find the total · '5 or 6': add their frequencies

ROUND 3 Questions 9–11: Vinh spun this spinner 80 times. Use the results.



Colour	Red	Green	Yellow
Frequency	44	25	11

9 Find the observed P(red).

Working _____

Answer _____

10 Find the theoretical P(red).

Working _____

Answer _____

11 Find the observed P(yellow).

Working _____

Answer _____

ROUND 4 Questions 12–14: the matches in 50 matchboxes were counted.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

12 How many matchboxes were tested? Check by adding.

Working _____

Answer _____

13 Find the observed P(exactly 51 matches).

Working _____

Answer _____

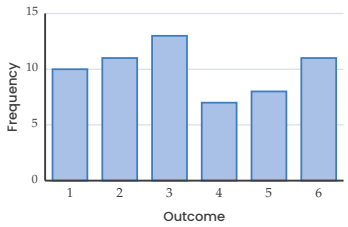
14 Find the observed P(fewer than 50 matches).

Working _____

Answer _____

ROUND 5 Questions 15–17: Declan rolled a die 60 times.

Outcome	1	2	3	4	5	6
Frequency	10	11	13	7	8	11



15 Find the observed P(3).

Working _____

Answer _____

16 Find the observed P(5 or 6).

Working _____

Answer _____

17 Find the theoretical P(5 or 6).

Working _____

Answer _____

ROUND 6 Questions 18–19: run your own simulation with a random number generator.

18 Use RanInt#(1,6) on your calculator (or =RANDBETWEEN(1,6) in a spreadsheet) to roll a die 30 times. Tally your results.

Outcome	1	2	3	4	5	6
Tally						
Frequency						

19 Find your observed P(6). Compare it with the theoretical $P(6) = \frac{1}{6}$. Why might they differ?



Answer _____

5 Observed probability

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Read each table carefully: the total number of trials is the bottom of the fraction.

Matches in 50 boxes

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

A die rolled 80 times

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

a Table
Matchboxes: find the observed P(exactly 51 matches).

b Fewer than
Matchboxes: find the observed P(fewer than 50 matches).

c Biased?
The die rolled 80 times: do you think it is biased? Give a reason.

d Simulate
Use RanInt#(1,6) to roll a die 20 times. Find your observed P(an even number).



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **5 QUESTIONS**

REMEMBER observed P = frequency ÷ total trials · more trials give a better estimate

1 Kai tosses a coin 200 times and gets 92 heads. Find the observed P(tails), as a decimal.

Answer

2 A pair of dice was rolled 50 times and the sum recorded. Find the observed P(a sum of 8), as a decimal.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

Answer

3 Use the table in Question 2. Find the observed P(a sum of 5 or 6).

Answer

4 Use the table in Question 2. Which sum came up most often? Find its observed probability.

Answer

5 Write $\frac{13}{50}$ as a percentage. Year 7

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1 Harper rolled a die 60 times and 2 came up 15 times. Harper says the die is biased. Is there enough evidence? What could Harper do to check?

Answer

2 The canteen's last 200 lunch orders were 70 wraps, 50 sushi and 80 burgers. Find the observed P(the next order is a wrap). Why is this only an estimate?

Answer

3 To estimate P(a family of 4 children are all girls), Jo tosses 4 coins (heads = girl) 50 times. 4 heads came up 4 times. Find the observed probability. How could Jo improve the estimate?



Answer

6 Expected frequency

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to calculate the expected frequency of an event, and compare it with the observed frequency.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find expected frequency = probability × number of trials. ○○○
- *I can* compare observed and expected frequencies. ○○○
- *I can* use an observed probability to make a prediction. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Expected frequency = probability × number of trials.

a Coin

A coin is tossed 50 times. Find the expected number of heads.



b Die

A die is rolled 72 times. Find the expected number of 4s.



c Spinner

Vinh spins 80 times. Find the expected number of reds. He got 44: compare.



d Rain

P(rain on an April day) = 30%. Find the expected number of rainy days in April.



YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER expected frequency = probability × number of trials, e.g. $\frac{1}{6} \times 60 = 10$

ROUND 1 Questions 1–4: find each expected frequency: probability × number of trials.

1 A coin is tossed 100 times. Expected number of heads?



P(E) _____
× trials _____
Expected _____

2 A die is rolled 60 times. Expected number of 6s?



P(E) _____
× trials _____
Expected _____

3 This spinner is spun 40 times. Expected number of blues?



P(E) _____
× trials _____
Expected _____

4 A die is rolled 30 times. Expected number of even numbers?



P(E) _____
× trials _____
Expected _____

ROUND 2 Questions 5–8: Vinh's spinner (half red, a quarter green, a quarter yellow) was spun 80 times: red 44, green 25, yellow 11.

5 Find the expected number of reds.

Working _____
Answer _____

6 Find the expected number of yellows.

Working _____
Answer _____

7 Red came up 44 times. Is that more or less than expected? By how many?

Working _____
Answer _____

8 Yellow came up 11 times. Compare it with the expected number.

Working _____
Answer _____

6 Expected frequency

Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 11 QUESTIONS

REMEMBER expected frequency = P × trials · from data: use the observed probability · it might not be a whole number

ROUND 3 — Questions 9–11: a fair die is rolled 72 times. Find each expected frequency.

9 Expected number of 4s

Working _____

Answer _____

10 Expected number greater than 2

Working _____

Answer _____

11 Expected number of even numbers

Working _____

Answer _____

ROUND 4 — Questions 12–14: use the observed probability to predict.

12 A basketballer scores 7 of every 10 shots. How many goals are expected from 50 shots?

Working _____

Answer _____

13 A surfer catches a wave on 3 of every 4 attempts. How many waves are expected from 60 attempts?

Working _____

Answer _____

14 20 of 50 sample matchboxes held exactly 50 matches. How many of 250 boxes would you expect to?

Working _____

Answer _____

ROUND 5 — Questions 15–17: Kingscliff weather records show that P(a summer day is above 30 °C) = 0.2.

15 How many days above 30 °C are expected in a 90-day summer?

Working _____

Answer _____

16 How many are expected in two weeks (14 days)?

Working _____

Answer _____

17 Give a reason why the real number of hot days might be different.

Working _____

Answer _____

ROUND 6 — Questions 18–19: a coin experiment: observed against expected.

18 Toss a coin 50 times (or use RanInt#(1,2), 1 = heads). Tally the results.

Outcome	Tally	Frequency
Heads		
Tails		
Total		50

19 What is the expected number of heads in 50 tosses? Compare it with your observed frequency.

Answer _____

6 Expected frequency

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Use theoretical probability when the outcomes are equally likely; otherwise use the observed probability.

a From data

Harper's die: 15 twos in 60 rolls. Using this, how many 2s are expected in 500 rolls?



b Matchboxes

20 of 50 sample boxes held exactly 50 matches. How many of 1000 boxes would you expect to?

c Not whole

A die is rolled 80 times. Find the expected number of 6s. It came up 22 times: comment.

d Lollies

P(red) = 0.5, P(green) = 0.2, P(purple) = 0.3. How many of each colour would you expect in a bag of 100?

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **5 QUESTIONS**

REMEMBER expected = P × trials · compare observed and expected: close suggests fair; far off over many trials suggests biased

1. A die was rolled 80 times. Find the expected number of 6s, to 1 decimal place. Compare it with the observed number.

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

Answer

2. Use Question 1. Do you think the die is biased? Explain.

Answer

3. A fair die is rolled 100 times. Find the expected number of 3s, to 1 decimal place.

Answer

4. Find 35% of 240. Year 7

Answer

5. 41 of 50 sample matchboxes held at least 50 matches. How many of 1000 boxes would you expect to?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1. Declan rolled a die 60 times: 1 (10), 2 (11), 3 (13), 4 (7), 5 (8), 6 (11). Find the expected and observed number of 5s or 6s, and comment.

Answer

2. A spinner has 5 equal sectors: 2 red, 2 blue, 1 green. Mel says she will get exactly 40 reds in 100 spins. Explain why she might not.

Answer

3. 8 students each toss a coin 100 times. How many heads are expected in total? The class got 389 heads. Comment.

Answer

CHAPTER SUMMARY

the key facts from every lesson, and how confident you are with each

HOW CONFIDENT AM I?

○○○ not yet · getting there · got it

1	Chance and the sample space	○○○	2	Probability of an event	○○○
3	Complementary events	○○○	4	Probability problems	○○○
5	Observed probability	○○○	6	Expected frequency	○○○

Key facts

Sample space: every possible outcome, each written once, in braces { } (L1)

$\mathbf{P(E)} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ when outcomes are equally likely (L2)

'or' for different groups: add the counts, e.g. P(red or blue) (L2)

Checklist: total → favourable → 'not'? → form asked → check (L4)

More trials: the observed probability usually gets closer to the theoretical one (L5)

Fair or biased? Close to expected: probably fair. Far off over many trials: probably biased (L6)

Probability scale: 0 impossible · $\frac{1}{2}$ even chance · 1 certain (L1)

Converting: fraction → decimal: top ÷ bottom; decimal → percentage: × 100% (L2)

Complement: P(not E) = 1 − P(E); all probabilities add to 1 (L3)

Observed P(E) $= \frac{\text{number of times E happened}}{\text{total number of trials}}$ (L5)

Expected frequency = probability × number of trials (L6)

IF YOU WERE AWAY

two examples from each lesson, fully worked: read them, then try the lesson's first round

Lesson 1 • Chance and the sample space

the lesson is on p. 2

Example 1: a sample space

List the sample space for a spinner with sectors red, blue, red, green.

Step 1 Write each colour once: {red, blue, green}.

Step 2 Count: 3 outcomes.

Step 3 Red has 2 of the 4 sectors, so the outcomes are **not equally likely**.

Example 2: chance words

The forecast says a 30% chance of rain. Describe the chance.

Step 1 Compare with 50% (even chance): 30% is less.

Step 2 So rain is **unlikely**.

Step 3 It is not 0%, so it is still possible.

Lesson 2 • Probability of an event

the lesson is on p. 5

Example 1: P(E) from counters

A bag has 5 red, 3 blue and 4 green counters. Find P(red).

Step 1 Total = 5 + 3 + 4 = 12.

Step 2 Favourable: 5 red.

Step 3 P(red) $= \frac{5}{12}$

Example 2: a percentage

A die is rolled. Find P(a number greater than 4), as a percentage.

Step 1 Favourable: {5, 6}, 2 of 6.

Step 2 $\frac{2}{6} = \frac{1}{3}$

Step 3 $\frac{1}{3} \times 100\% = 33\frac{1}{3}\% \approx 33.3\%$

Lesson 3 • Complementary events

the lesson is on p. 8

Example 1: P(not E) with decimals

$P(\text{rain}) = 0.15$. Find $P(\text{no rain})$.

Step 1 'No rain' is the complement of 'rain'.

Step 2 $\text{P}(\text{no rain}) = 1 - 0.15 = \mathbf{0.85}$

Example 2: P(not E) with fractions

A pack has 10 red, 6 yellow and 4 green cards. Find $P(\text{not yellow})$.

Step 1 $\text{P}(\text{yellow}) = \frac{6}{20} = \frac{3}{10}$

Step 2 $\text{P}(\text{not yellow}) = 1 - \frac{3}{10} = \mathbf{\frac{7}{10}}$

Lesson 4 • Probability problems

the lesson is on p. 11

Example 1: an 'or' problem

A number from 1 to 15 is chosen. Find $P(\text{a multiple of 3 or 5})$.

Step 1 List: $\{3, 5, 6, 9, 10, 12, 15\}$ (15 once).

Step 2 7 of 15: $\text{P} = \mathbf{\frac{7}{15}}$

Example 2: no memory

A coin has shown tails 7 times in a row. Find $P(\text{tails next})$.

Each toss is a fresh start: the coin does not remember.

$\text{P}(\text{tails}) = \mathbf{\frac{1}{2}}$

Lesson 5 • Observed probability

the lesson is on p. 14

Example 1: observed probability

A die is rolled 60 times; 2 comes up 15 times. Find the observed $P(2)$.

Step 1 Observed P = times ÷ trials $= \frac{15}{60} = \mathbf{\frac{1}{4}}$

Step 2 Theoretical $\text{P}(2) = \frac{1}{6}$, so 2 came up more than expected.

Example 2: from a table

Matches in 50 boxes: 48 (3), 49 (6), 50 (20), 51 (16), 52 (4), 53 (1). Find the observed $P(\text{fewer than } 50)$.

Step 1 Fewer than 50: 48 or 49, $3 + 6 = 9$ boxes.

Step 2 $\text{P} = \mathbf{\frac{9}{50}}$

Lesson 6 • Expected frequency

the lesson is on p. 17

Example 1: expected frequency

A die is rolled 72 times. Find the expected number of 4s.

Expected = P × trials $= \frac{1}{6} \times 72 = \mathbf{12}$

Example 2: predicting from data

Harper's die gave 15 twos in 60 rolls. How many 2s in 500 rolls?

Step 1 Observed $\text{P}(2) = \frac{15}{60} = \frac{1}{4}$

Step 2 $\frac{1}{4} \times 500 = \mathbf{125}$

R1 [L1] List the sample space for a spinner with sectors 1, 2, 2 and 3. How many outcomes are there?

R2 [L1] A drawing pin is dropped: it lands point up or point down. Are the outcomes equally likely? Explain.

R3 [L2] A bag has 6 red, 4 blue and 2 white marbles. Find P(blue).

R4 [L2] A die is rolled. Find P(a number less than 5), as a percentage.

R5 [L2] One card is taken from a standard deck. Find P(a black king).

R6 [L2] Write $\frac{7}{20}$ as a decimal and as a percentage.

R7 [L3] Write the complement of 'Kingscliff wins the final'.

R8 [L3] P(the bus is late) = 0.12. Find P(the bus is not late).

R9 [L3] A spinner: P(red) = 0.45, P(blue) = 0.3 and the rest is green. Find P(green).

R10 [L3] A family of 5 (2 parents, 3 children) take turns to wash up. Find P(the washer-up is not a parent).

R11 [L4] One tile is chosen from T, W, E, E, D. Find P(an E), as a decimal.

R12 [L4] The digits 1, 3, 5 and 8 make a 4-digit number at random, each used once. Find P(the number is odd).

R13 [L5] A coin is tossed 80 times and lands heads 36 times. Find the observed P(heads), as a decimal.

R14 [L5] Theoretical or observed? 'P(a Year 8 student surfs) = 0.3, from a survey.' Explain.

R15 [L6] A die is rolled 90 times. Find the expected number of 5s.

R16 [L6] A basketballer scored 30 of 50 shots. Predict the number scored in 120 shots.

EXIT TICKET Lesson 1: Chance and the sample space Name _____ Class _____

Easy • List the sample space for a spinner with sectors red, blue, red.

Medium •• Is each colour on that spinner equally likely? Explain.

Hard ••• Describe, in words, the chance of rolling less than 3 on a die.

EXIT TICKET Lesson 2: Probability of an event Name _____ Class _____

Easy • A die is rolled. Find P(a 4).

Medium •• A bag has 3 red and 5 blue counters. Find P(blue), as a percentage.

Hard ••• One card from a deck. Find P(a red ace).

EXIT TICKET Lesson 3: Complementary events Name _____ Class _____

Easy • Write the complement of 'rolling an even number'.

Medium •• P(win) = 0.35. Find P(not win).

Hard ••• A jar has 7 red, 5 green and 8 yellow lollies. Find P(not yellow).

EXIT TICKET Lesson 4: Probability problems Name _____ Class _____

Easy • A spinner has 5 equal sectors, 1 to 5. Find P(odd), as a percentage.

Medium •• 30 students: 12 bus, 10 walk, the rest are driven. Find P(driven).

Hard ••• A coin lands heads 5 times in a row. Find P(heads next). Explain.

✂

SUMMARISE LESSON 1 IN YOUR OWN WORDS

Chance and the sample space • the back of your exit ticket

✂

SUMMARISE LESSON 2 IN YOUR OWN WORDS

Probability of an event • the back of your exit ticket

✂

SUMMARISE LESSON 3 IN YOUR OWN WORDS

Complementary events • the back of your exit ticket

✂

SUMMARISE LESSON 4 IN YOUR OWN WORDS

Probability problems • the back of your exit ticket

✂

✂ - - -

✂

SUMMARISE LESSON 5 IN YOUR OWN WORDS

Observed probability • the back of your exit ticket

✂

SUMMARISE LESSON 6 IN YOUR OWN WORDS

Expected frequency • the back of your exit ticket

✂

HOMEWORK STRIPS — after each lesson, cut out its strip (start at the top right), do it at home, and hand it in next lesson

HW 7 Before the test 1 ✂

1 **L1** How many outcomes are in the sample space for the day of the week?

2 **L2** A 10-sided die (1 to 10). Find P(an even number).

3 **L3** A die is rolled. Find P(not a 3).

HW 4 L4 Problems ✂

1 A number from 1 to 10 is chosen. Find P(a prime).

2 One tile from T, W, E, E, D. Find P(not an E).

3 A bag of 20 lollies has P(red) = 0.5. How many are red?

HW 1 L1 Chance ✂

1 List the sample space for tossing a coin.

2 Choose a chance word for a probability of 0.8.

3 Are the outcomes of rolling a fair die equally likely?

HW 8 Before the test 2 ✂

1 **L4** A drawer has 3 black socks and 1 white sock. Find P(white).

2 **L5** 12 of 40 spins were blue. Find the observed P(blue).

3 **L6** P(E) = 0.3. Find the expected number in 50 trials.

HW 5 L5 Observed ✂

1 50 tosses gave 27 heads. Find the observed P(heads).

2 Theoretical or observed: 'P(heads) = ½'?

3 Why is it better to do many trials?

HW 2 L2 P(E) ✂

1 A die is rolled. Find P(a number greater than 2).

2 A bag has 2 red and 6 blue counters. Find P(red), as a decimal.

3 One card from a deck. Find P(a spade).

HW 9 Before the test 3 ✂

1 **L2** One card from a deck. Find P(an ace).

2 **L3** P(win) = 65%. Find P(not win).

3 **L6** 60 rolls gave 25 sixes. Is the die biased?

HW 6 L6 Expected ✂

1 Find the expected number of 6s in 300 rolls of a die.

2 Find the expected number of heads in 70 coin tosses.

3 A spinner is half red. Find the expected number of reds in 40 spins.

HW 3 L3 Complements ✂

1 P(rain) = 0.4. Find P(no rain).

2 Write the complement of 'winning a raffle'.

3 One tile from N, I, P, P, E, R, S. Find P(not a vowel).

HW 1 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 2 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 3 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 4 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 5 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 6 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 7 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 8 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 9 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working: